

Notations :-

IPv4 = 32 bits(4 bytes)

IPv6 = 128 bits(16 bytes)

2^0

2^1

2^2

.

.

.

.

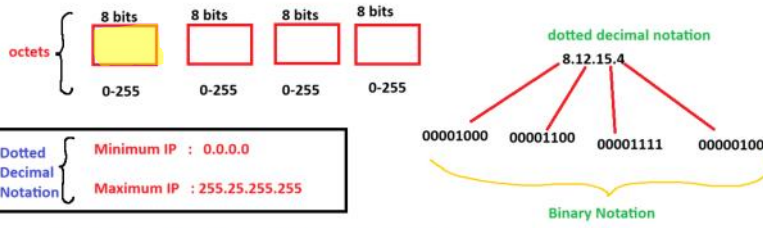
.

.

.

2^{31}

$2^8 = 256(0-255)$

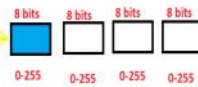


Dotted Decimal Notation
Minimum IP : 0.0.0.0
Maximum IP : 255.255.255.255

Binary Notation
Minimum IP : 00000000 00000000 00000000 00000000
Maximum IP : 11111111 11111111 11111111 11111111

Classful Addressing :-

First octet (first byte) of any IP address represents the CLASS



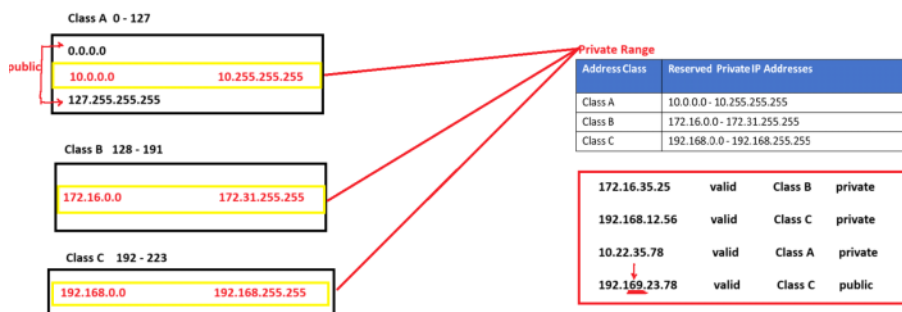
Class A 0 - 127
Class B 128 - 191
Class C 192 - 223
Class D 224 - 239
Class E 240 - 255

0.0.0.0 to 127.255.255.255
128.0.0.0 to 191.255.255.255
192.0.0.0 to 223.255.255.255
224.0.0.0 to 239.255.255.255
240.0.0.0 to 255.255.255.255

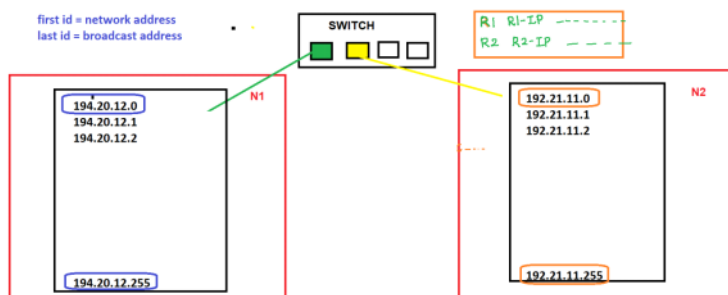
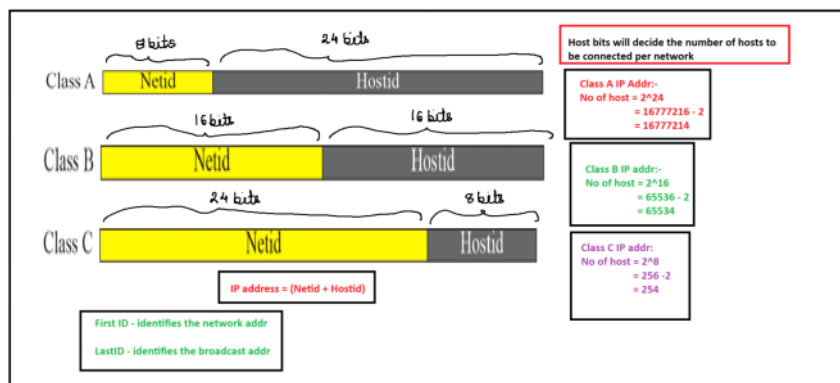
120.13.45.67 valid Class A
193.23.47.89 valid Class C
246.13.24.65 valid Class E
X 129.12.67.266 invalid
0-255

0	10	110	1110	1111
Class A 0 - 127	Class B 128 - 191	Class C 192 - 223	Class D 224 - 239	Class E 240 - 255
0 0000 0000 1 0000 0001 2 0000 0010 ... 127 0111 1111	128 1000 0000 129 1000 0001 ... 191 1011 1111	192 1100 0000 193 1100 0001 ... 223 1101 1111	224 1110 0000 225 1110 0001 ... 239 1110 1111	240 1111 0000 241 1111 0001 ... 255 1111 1111

Private IP addresses :-



NetId, HostId of A,B,C classes :-



7 layers of OSI Model :-

